Replication of Card and Krueger (1994)

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1. Research Question

Card and Krueger (1994) study whether a higher minimum wage reduces employment. In April 1992, New Jersey increased its minimum wage from $4.25 to $5.05, while Pennsylvania kept its minimum wage at $4.25.

A higher minimum wage raises labor costs, which could lead restaurants to hire fewer workers. The authors test this by comparing employment changes in fast-food restaurants in the two states. They find no evidence that employment fell in New Jersey relative to Pennsylvania.

This project replicates Tables 2–4 and Figure 1. It also checks whether excluding permanently closed restaurants changes the main employment result.

2. Data and Variables

I used public.csv from the course data package. The dataset contains 410 restaurants: 331 in New Jersey and 79 in Pennsylvania. Restaurants were surveyed before and after the minimum-wage increase.

Employment is measured in full-time equivalents:

FTE = full-time employees + managers + 0.5 × part-time employees.

If an employment component is missing, FTE is treated as missing. Permanently closed restaurants have post-policy employment set to zero. Temporarily closed restaurants are excluded from the baseline employment comparison.

Different analyses use different samples. The balanced sample includes 384 restaurants with valid employment in both periods. Table 4 uses 357 restaurants after applying the additional wage-data requirements. This sample includes permanent closures even when their post-policy wages are missing.

3. Method

The main method is difference-in-differences:

DiD = employment change in NJ − employment change in PA.

Pennsylvania provides a comparison for what might have happened in New Jersey without the wage increase. This requires the assumption that employment in the two states would otherwise have followed similar trends. The data contain only one pre-policy survey, so this assumption cannot be tested using earlier trends.

Table 4 also uses regressions to account for restaurant characteristics. Some models use an NJ indicator; others use GAP, which measures how much a restaurant needed to raise its starting wage to reach $5.05.

For NJ restaurants:

GAP = max[0, (5.05 − starting wage before the policy) / starting wage before the policy].

GAP is zero for PA restaurants. The regressions use conventional OLS standard errors.

4. Replication Results

Table 2 reproduces many of the paper's descriptive results. Average starting wages in NJ rose from $4.61 to $5.08. In PA, they remained close to $4.63 before the policy and $4.62 afterward. Figure 1 shows that NJ wages became concentrated around the new minimum wage.

The main employment results are:

Sample	NJ employment change	PA employment change	DiD	SE
Balanced sample	0.47	−2.28	2.75	1.34
Temporary closures assigned zero employment	0.23	−2.28	2.51	1.35

In the balanced sample, NJ employment increased slightly while PA employment declined. The resulting DiD is **2.75 FTE workers per restaurant**, matching the paper's balanced-sample estimate. This is a relative change between the states, not the number of workers added by every NJ restaurant.

Table 4 gives the following results using 357 restaurants:

Model	Main variable	Estimated coefficient	SE	Paper coefficient
(i)	NJ indicator	2.33	1.19	2.33
(ii)	NJ indicator	2.30	1.20	2.30
(iii)	GAP	15.65	6.08	15.65

(iv)	GAP	14.92	6.21	14.92
(v)	GAP	11.98	7.42	11.91

Models (ii) and (iv) include chain and ownership controls. Model (v) adds regional controls. The first four main coefficients match the paper at its reported precision. Model (v) differs slightly, and the reason remains unclear. GAP is a proportional measure, so its coefficient should not be interpreted as the effect of a one-dollar wage increase.

The complete tables are provided in the outputs/ folder.

5. Extension: Excluding Permanently Closed Restaurants

I examined whether the employment result changes when permanently closed restaurants are removed. This excludes five NJ restaurants and one PA restaurant from the balanced sample.

Sample	NJ restaurants	PA restaurants	DiD	SE
Including permanent closures	309	75	2.7500	1.3423
Excluding permanent closures	304	74	2.7219	1.3401

The estimate changes from 2.75 to 2.72, a decrease of about 0.03 FTE workers. The result is therefore similar under this sample change.

However, excluding closed restaurants leaves out the jobs lost through closure. This check supplements the main analysis rather than replacing it. Since the two samples overlap, I do not treat the estimates as independent when comparing them.

6. Validation and Limitations

I compared the outputs with the original tables and checked the sample sizes. Several differences remain.

The full-time employee percentage in Table 2 has not been reproduced because its exact definition is unresolved. The NJ post-policy share earning exactly $5.05 is 85.50%, compared with 85.2% in the paper. Some t values also differ. The use of SPECIAL2 for the post-policy recruiting-bonus measure still needs confirmation.

The standard errors in Table 3's third row have not been reproduced and are marked as NA. Its NJ−PA estimate rounds to 2.75, compared with 2.76 in the paper.

More generally, differences in economic conditions between NJ and PA could affect the employment comparison. The reported standard errors also do not address possible common shocks within each state. The results should therefore be interpreted cautiously.

7. Reproducibility and Conclusion

After placing the input file at data/public.csv, the command python3 run_all.py generates the analysis outputs and processed data. The README explains the setup, and Git records the main stages of the project. AI use is described separately in AI_USE.md.

The replication produces results close to the paper's main employment findings. NJ employment increased relative to PA, and excluding permanent closures makes little difference to the estimate. This supports a similar conclusion for this sample and period, but it does not show that every minimum-wage increase will raise employment.

Reference

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review*, 84(4), 772–793.